

Song Matching Analysis

People are constantly looking for new music to listen to. Modern music aggregation apps such as SPotify have made access to music easier than ever, and to keep people coming back, they need to have ways to introduce listeners to new songs they may like. This also helps smaller artists who may be less known to get their music to audiences that might enjoy it based on their tastes. There are many different methods to identify songs that are most similar to each other. A common, fairly intuitive way is to analyze word usage. The first method we used in this examination was the raw inner product. We used this to compare songs based on the words they have in common and the frequency of those words within the songs. However, one issue with this method is that it can be biased toward longer songs that contain a larger number of words, since songs with more words will generally produce larger inner product values and more chances for the words to match. This led to some seriously skewed results. To address this issue, cosine similarity was used next, which focused more on the direction of the song vectors rather than just their overall size. This makes the comparison less dependent on the total number of words in a song, which is what we were looking for. Even still, this method is comparing every word, and some words like prepositions and pronouns aren't going to be useful for matching similar songs. Words like "the" and "they" are said so much in every song that they could skew results. To further combat this issue, TF-IDF was incorporated into the cosine similarity calculation to give different weights to words based on how common or rare they are across the entire collection of songs. This allowed words that are more unique and potentially more important to a song to have a greater influence on the similarity score.

We then evaluated the models based on how often they recommended songs that were from the same genre as the original song. Surprisingly, the initial methods were not particularly strong at matching songs within the same genre. The raw inner product produced a same-genre match only about 12% of the time, while the cosine similarity methods produced same-genre matches only about 6% of the time. This suggests that similarities in word usage alone do not necessarily translate into similarities in genre, which does make sense. Genre has more to do with sound, tempo, and rhythm a lot of the time as opposed to just what words are said. To improve the recommendations, we incorporated genre directly into the similarity calculation by assigning it a 20% weight in the overall score. This produced a dramatic improvement in the genre matching part of the analysis, with the resulting recommendations matching the original song's genre 96% of the time. This shows that while the word-based similarity methods can identify songs with similar lyrics and expressed words, incorporating genre information can make the recommendations much more effective at producing songs from the same genre. This may be a better route for music sites to recommend songs to listeners. It makes sense because if you are enjoying a song from the blues genre, it wouldn't make much sense to recommend a song from hip-hop just because it says similar words. This way, you stay within the same genre, while also finding songs with similar sound and lyrics alike.